AI-Powered Skin Intelligence & Personalized Skincare Routine Recommendation Platform

Comprehensive Project Technical Report

Program: Infosys Springboard Digital Health Capstone **Target Audience:** Engineering Team, Evaluators, and Clinical Stakeholders **Date:** October 2026 **Status:** Milestone 4 Complete, Production Ready

1. Introduction

Skin is the human body's largest organ and its primary barrier against environmental pathogens, ultraviolet radiation, and toxic chemicals. Yet personal skincare remains one of the most unregulated and empirically misunderstood consumer domains. Consumers routinely make high-stakes chemical choices based on marketing claims, influencer endorsements, and unscientific product claims.

Active agents such as Retinoids, Alpha-Hydroxy Acids (AHAs), Beta-Hydroxy Acids (BHAs), L-Ascorbic Acid (Vitamin C), and Niacinamide can dramatically improve dermatological outcomes when used at correct concentrations, at the right times, and with complementary actives. Combined haphazardly, they trigger severe trans-epidermal water loss (TEWL), lipid matrix breakdown, and acute chemical dermatitis.

Dermatologists also recognize skin as an endocrine-responsive organ. Its health depends not only on topical products but on circadian sleep cycles, systemic hydration, psychological cortisol elevation, and cumulative UV exposure. Despite this, standard consumer applications treat skincare as a static questionnaire. They offer no real-time adaptation, no protection against dangerous chemical combinations, no predictive modeling of habit changes, and no direct integration with certified medical professionals.

The AI-Powered Skin Intelligence Platform was engineered to resolve this gap. It bridges consumer skincare and clinical dermatology through an enterprise-grade digital health ecosystem built on predictive Machine Learning models, natural language formulation vectorization, deterministic cosmetic chemistry matrices, and circadian protocol scheduling.

2. Abstract

This project presents an enterprise-grade, full-stack digital health application that delivers clinically guided, biologically synchronized skincare personalization and risk mitigation. The platform combines four machine learning and AI capabilities:

- **Continuous Regressors:** Random Forest ensembles trained to predict a dynamic Skin Health Index (0 to 100) and rank skin concern priorities from daily lifestyle telemetry.
- **Natural Language Processing (NLP):** TF-IDF vectorization with Cosine Similarity to analyze International Nomenclature of Cosmetic Ingredients (INCI) formulations and match them to patient profiles.
- **Deterministic Expert System:** A dermatological Cosmetic Chemistry Matrix that intercepts dangerous active ingredient clashes (for example, Retinoids with exfoliating acids) and flags synergistic combinations in real time.
- **Generative AI:** Google Gemini 2.5 Flash, used under strict medical guardrails to turn longitudinal telemetry into formal clinical progress summaries.

The system is a containerized, 3-tier microservice application using Next.js 16 (React 19) with Turbopack, FastAPI on Python 3.11, and PostgreSQL 15 managed through SQLAlchemy 2.0. It supports dual role-based workflows for general users and licensed dermatologists, and includes an interactive counterfactual "What-If" habit simulator, a 5-point polar-to-Cartesian radar chart, exportable medical PDF reports, and an automated test framework of 48 passing unit and integration tests (100% pass rate). Cloud deployments are active on Vercel (frontend), Render (backend API), and Supabase (PostgreSQL database).

3. Problem Statement

Modern dermatological health and consumer skincare face five acute, industry-wide challenges.

3.1 Active Ingredient Conflict and Chemical Erythema. Access to high-potency actives has surged without matching consumer chemical literacy. Combining incompatible compounds, such as low-pH L-Ascorbic Acid with Niacinamide or Retinol with Glycolic Acid, causes acute barrier disruption, persistent contact dermatitis, and micro-scarring.

3.2 Static Recommendations in a Dynamic Biological System. Standard applications rely on one-time onboarding quizzes that recommend the same routine regardless of environmental or physiological change. In reality, skin undergoes continuous cellular turnover (about 28 days for human keratinocytes) and fluctuates daily with sleep deprivation, dehydration, and stress.

3.3 The Counterfactual Blind Spot. Patients struggle to understand why their skin is deteriorating and which habit has the greatest negative leverage. Without mathematical attribution, they cannot tell whether two more hours of sleep or four more glasses of water would improve their skin faster.

3.4 The Clinician-Patient Telemetry Disconnect. Dermatologists spend consultation time collecting subjective, unreliable verbal histories about products used, timing, and habit changes over preceding months. Without an objective, longitudinal daily record, accurate diagnosis and treatment modification suffer.

3.5 Lack of Automated Testing and Clinical Guardrails in Health Tech. Many commercial health apps lack rigorous input sanitization, schema validation, and boundary verification. This leads to runtime crashes, data corruption, or unsafe recommendations when users enter non-standard inputs.

4. Proposed Solution

The platform addresses these challenges with an end-to-end, scientifically grounded ecosystem:

- **Automated Chemical Safety Enforcement:** An INCI Cosmetic Chemistry Clash Engine parses product formulations and user-selected serums, flagging contraindications, pH clashes, and barrier damage warnings before application.
- **Circadian-Synchronized Protocol Generation:** A routine scheduler splits regimens into Morning AM Protection (antioxidants, hydration barriers, broad-spectrum SPF 50) and Evening PM Cellular Repair (gentle cleansing, cell turnover actives, lipid barrier restoration).
- **Predictive ML and What-If Simulation:** A Random Forest regressor computes dynamic health indices, while the simulator projects 7-day, 14-day, and 30-day trajectories under counterfactual habits, with Gini feature importance ranking.
- **Exportable Medical Reports with 5-Point Radar:** A visualizer maps five biological dimensions (Hydration, Barrier, Oil, Sun, Sensitivity) against healthy population benchmarks, with a Generative AI clinical sign-off and one-click A4 PDF export.
- **Clinician Command Center:** A workspace for certified dermatologists with patient roster management, ML concern priority triage, daily telemetry audits, and persistent clinical notes.
- **Hardened Enterprise Architecture:** A 3-tier Docker Compose stack backed by a 48-test PyTest suite (100% pass rate), with zero database side-effects and strict Pydantic V2 validation.

5. Technology Stack

5.1 Frontend Tier

- **Framework:** Next.js 16.3.1 with the App Router, React 19 Server and Client Components, and the Turbopack bundler.

- **Language:** TypeScript with strict type safety, defensive memoized data normalizers, and custom type definitions.
- **UI and Styling:** Tailwind CSS for responsive design, adaptive Dark/Light theming, and Lucide React icons.
- **Visual Math and Export:** Native SVG polygon trigonometry for the 5-point radar, with CSS `@media print` rules for A4 medical PDF output.

5.2 Backend Tier

- **Framework:** FastAPI, asynchronous, on Python 3.11 with the Uvicorn ASGI server.
- **Validation and Security:** Pydantic V2 models enforcing strict biological and numerical boundaries, passlib bcrypt password hashing, and OAuth2 JSON Web Tokens (JWT) for stateless authentication.
- **Testing:** PyTest, the HTTPX asynchronous test client, and an isolated in-memory SQLite test database.

5.3 Machine Learning and Data Science Tier

- **Frameworks:** Scikit-learn, NumPy, Pandas, and Joblib for model loading and array computation.
- **Predictive Models:** Pre-trained Random Forest continuous and multi-output regressors (`skin_score_model.pkl` , `concern_priority_model.pkl` , `adherence_model.pkl` , `ingredient_safety_model.pkl`).
- **NLP Pipeline:** Scikit-learn `TfidfVectorizer` with pairwise `cosine_similarity` .
- **Generative AI:** Google GenAI SDK with Gemini 2.5 Flash and clinical system prompt guardrails.

5.4 Database and DevOps Infrastructure

- **Database:** PostgreSQL 15 with automated connection pooling and schema management through SQLAlchemy 2.0.
- **Containerization:** `docker-compose.yml` orchestrating three isolated services (`skin_intelligence_db` , `skin_intelligence_backend` , `skin_intelligence_frontend`) on an internal bridge network.
- **Cloud Hosting:** Vercel (frontend), Render (FastAPI backend), and Supabase (PostgreSQL).

6. Features in Detail

Feature 1: User Authentication and Role-Based Access Control (RBAC)

Purpose: Secure, segregated access between general patients and certified dermatologists, protecting health records from unauthorized access.

User workflow: Patients register at `/register` and log in at `/login`. Clinicians use a dedicated entry point at `/login/dermatologist`. Navigation, permissions, and dashboards adapt to the authenticated token.

Engineering: The backend implements the OAuth2 Password Bearer flow in FastAPI. Passwords are irreversibly hashed with bcrypt and adaptive salt rounds. JWT tokens carry `sub` (email), `user_id`, and `role` claims with a 24-hour expiry. Endpoints enforce access through the dependencies `security.get_current_user` and `security.require_dermatologist`.

Feature 2: Smart Clinical Skin Diagnostic and Profiling Engine

Purpose: Establish an accurate biological baseline covering skin type, barrier reactivity, and specific concerns to guide every AI recommendation.

User workflow: Users complete a clinical onboarding questionnaire at `/dashboard/profile`. They select a skin category (Dry, Normal, Combination, Oily, Sensitive), primary concerns (Acne Vulgaris, Hyperpigmentation, Photo-Aging, Dehydration, Redness, Dullness), and barrier sensitivity flags (reactive skin, burning sensation, eczema history).

Engineering: The payload is validated by the Pydantic V2 `SkinProfileCreate` schema and stored in the `skin_profiles` table with a foreign key to `users.id`. Saving triggers the initial baseline skin score calculation and seeds the user's first routine.

Feature 3: Circadian AM/PM Regimen Generator

Purpose: Remove confusion about product order and align steps with human circadian cycles (daytime protection, nighttime restoration).

User workflow: At `/dashboard/routine`, users see two schedules:

- **Morning AM Protocol (Protection):** gentle low-pH cleansing, antioxidant serums (Vitamin C/E), lightweight barrier hydration, and broad-spectrum SPF 50.
- **Evening PM Protocol (Cellular Repair):** oil-based double cleansing, targeted actives (Retinol, BHA, Azelaic acid), and rich ceramide barrier repair cream.

Each step shows instructions, wait times, and target biological actions.

Engineering: A dermatological decision tree enforces three rules. The *Photosensitivity Rule* locks actives that degrade in UV or raise photo-reactivity (Retinoids, AHAs) to the PM routine. The *Photoprotection Rule* maps free-radical scavengers and SPF agents strictly to the AM routine. The *Sensitivity Guardrail* throttles active frequency to alternate evenings for sensitive skin.

Feature 4: Content-Based Skincare Formulation Recommender

Purpose: Replace sponsored advertising with objective, algorithmic recommendations based on formulation compatibility.

User workflow: At `/dashboard/products`, curated product cards show dynamic Match Percentages (for example, 94% Match for Combination Acne-Prone Skin), along with benefits, active ingredients, and suitability rationale.

Engineering: The engine applies TF-IDF vectorization and Cosine Similarity to INCI ingredient strings across thousands of commercial products. It builds a normalized feature vector from the user's skin type, concern matrix, and sensitivity index, and scores each product with:

$$\text{Cosine Similarity}(u, p) = (u \cdot p) / (\|u\| \, \|p\|)$$

Formulations containing known allergens or harsh surfactants are filtered out before products are ranked by clinical match score.

Feature 5: INCI Cosmetic Chemistry and Active Clash Inspector

Purpose: Act as an automated personal cosmetic chemist, preventing users from layering incompatible actives that cause chemical burns and barrier degradation.

User workflow: At `/dashboard/ingredients`, users select active pairs or paste raw ingredient text. The inspector shows instant safety badges:

- **Dangerous Clash (red):** for example, Retinol + Glycolic Acid, with over-exfoliation risk, severe barrier stripping, and chemical erythema.
- **Caution / Moderate (yellow):** for example, L-Ascorbic Acid + Niacinamide, with potential flushing and pH neutralization.
- **Synergistic Pair (green):** for example, Hyaluronic Acid + Ceramides, giving optimal trans-epidermal hydration and lipid matrix lock.

Engineering: A deterministic Cosmetic Chemistry Rule Matrix derived from clinical dermatological literature scans pairwise active combinations, evaluates functional group compatibility, checks acidic and basic pH ranges, and issues real-time warnings.

Feature 6: Daily Lifestyle and Biomarker Telemetry Tracker

Purpose: Capture daily environmental, physiological, and behavioral inputs in under 30 seconds to feed the prediction models.

User workflow: At `/dashboard/tracker`, users record:

Input	Range / Options
Sleep duration	0 to 12+ hours (slider)
Water consumption	0 to 15+ glasses (counter)
Psychological stress index	1 (Relaxed) to 10 (Extreme)
Direct UV exposure	Hours in direct sunlight
Subjective barrier status	Normal, Dry/Flaking, Redness, Breakouts

An Adherence Streak Counter increments with consecutive daily logs.

Engineering: Pydantic V2 (`DailyTrackerCreate`) enforces strict boundaries. Transactional records are written to the `daily_logs` table, and the user's Skin Health Index and historical rolling averages are recalculated automatically.

Feature 7: Longitudinal Progress Tracking and Historical Analytics

Purpose: Give patients objective visual confirmation of their recovery trajectory, countering the discouragement common during slow 28-day turnover cycles.

User workflow: At `/dashboard/history` and `/dashboard/analytics`, interactive time-series charts show health score trends over 7, 14, and 30-day windows, with breakout frequency and redness severity histograms correlated against sleep and hydration logs.

Engineering: Optimized SQLAlchemy aggregation queries feed 7-day rolling moving averages that filter out transient daily variance. Recovery is modeled against human keratinocyte turnover timelines.

Feature 8: Executive Analytics and Counterfactual "What-If" Simulator

Purpose: Let users and clinicians simulate the effect of prospective lifestyle changes before adopting them, answering questions such as "What happens if I sleep 8 hours and drink 10 glasses of water?"

User workflow: At `/dashboard/executive`, sliders control sleep, water intake, and stress. The UI responds in real time (under 50 ms) with:

- **Projected Score Trajectory** over 7, 14, and 30 days.

- **Dynamic Risk Dials** for Dehydration, Barrier Stress, and Photo-Damage risk.
- **Gini Feature Importance Cards** showing the percentage weight of each driver (for example, Sleep Deficit: 38% impact, Chronic Stress: 27% impact).

Engineering: A pre-warmed Random Forest Continuous Regressor (`skin_score_model.pkl`) computes the counterfactual delta:

$\Delta = f(x_{simulated}) - f(x_{current})$

Gini impurity reduction weights are extracted directly from the trained tree ensemble to quantify feature attribution.

Feature 9: Clinical Health Reports and 5-Point Skin Balance Radar

Purpose: Convert telemetry logs into an exportable, medical-grade document suitable for hospital consultations.

User workflow: At `/dashboard/reports`, an interactive 5-point SVG radar covers five pillars: Hydration Balance (cellular moisture retention), Barrier Resilience (lipid matrix shield), Oil Control (sebum balance), Sun Defense (photoprotection index), and Sensitivity Defense (reactivity threshold). The patient's polygon is superimposed over a healthy population benchmark of 70.0. The page also provides an automated Generative AI Clinical Sign-Off and a "Print / Save as PDF" button.

Engineering: Polar-to-Cartesian mapping places each vertex i (i = 0 to 4):

$x_i = x_{center} + r_i \cdot \sin(2\pi i / 5)$, $y_i = y_{center} - r_i \cdot \cos(2\pi i / 5)$

The report connects to Gemini 2.5 Flash with clinical guardrails for a formal medical synthesis. Custom `@media print` CSS strips navigation and web controls to produce a clean A4 PDF.

Feature 10: Clinician / Licensed Dermatologist Command Center

Purpose: Give certified dermatologists a workspace to audit patient telemetry, review ML urgency triage, and document prescriptions.

User workflow: At `/dashboard/dermatologist`, a real-time roster shows risk tier badges (High Risk, Moderate Risk, Stable) with search and filters by email, skin type, or concern. "Open Profile" opens a clinical dossier modal containing:

- 5 biological radar pillars with cohort comparisons.
- Machine Learning Concern Priority Triage (Random Forest urgency ranking).
- A chronological telemetry table (sleep, water, stress, barrier feel).
- Morning and evening regimens with matched formulation actives.
- A Doctor Notes and Prescription Editor with a "Save Notes" action.

Engineering: Access is secured by `security.require_dermatologist`. `concern_priority_model.pkl` (Random Forest Multi-Output Regressor) calculates urgency scores and identifies driving telemetry factors. Notes persist transactionally through `POST /dermatologist/patient/{id}/notes`.

Feature 11: Automated Verification, QA and Schema Validation Suite

Purpose: Ensure that all ML models, recommendation engines, boundary checks, and API endpoints run without runtime errors.

Structure: The master runner is `python run_tests.py`, using PyTest and FastAPI's TestClient. All 48 of 48 tests pass in under 6 seconds. An isolated `sqlite:///:memory:` database guarantees no contamination of production data.

Test Suite	Tests	Coverage
`test_ml_models.py`	6	Deserialization, array shapes, and predictive consistency across all four `.pkl` artifacts
`test_engines.py`	12	TF-IDF cosine product matching, INCI clash matrix detection, radar coordinate geometry
`test_validations.py`	11	Pydantic V2 boundary checks; negative sleep, impossible water intake, and weak passwords return 422 Unprocessable Entity
`test_api_endpoints.py`	19	End-to-end tests for authentication, skin profiling, daily logs, routines, simulator, and reports

Feature 12: 3-Tier Containerization and Cloud Deployment

Purpose: Enable zero-downtime, reproducible local and cloud deployment across hosting environments.

Docker Compose orchestrates three containers:

- `skin_intelligence_db`: PostgreSQL 15 Alpine with a persistent `postgres_data` volume and `pg_isready -U postgres` healthcheck probes.
- `skin_intelligence_backend`: Python 3.11-slim, compiling native C dependencies, loading pre-warmed ML models, and serving FastAPI via Uvicorn on port 8001.
- `skin_intelligence_frontend`: Node 20-alpine running Next.js 16 with an optimized Turbopack build on port 3000.

Production cloud environments:

- **Frontend:** Vercel, with global edge CDN routing and automated CI/CD git integration.

- **Backend:** Render (`https://skin-intelligence-api.onrender.com`), Python 3.11 with SSL termination.
- **Database:** Supabase PostgreSQL with automated connection pooling.

7. Conclusion

The AI-Powered Skin Intelligence and Personalized Skincare Routine Recommendation Platform moves consumer skincare from fragmented, empirical advice to an integrated, scientifically validated digital health platform. By combining predictive Machine Learning ensembles, natural language formulation vectorization, deterministic cosmetic chemistry safety matrices, and circadian protocol scheduling, it addresses the core challenges of modern skincare:

- It safeguards patients against chemical burns and barrier destruction through automated clash interception.
- It accounts for real-world physiological dynamics by tracking daily sleep, hydration, and stress telemetry.
- It empowers patients through counterfactual "What-If" simulations that show the tangible benefits of healthy habits.
- It gives certified dermatologists objective, longitudinal biomarker data, streamlining consultations and improving patient outcomes.

Backed by a 48-test automated QA suite (100% pass rate), complete multi-container Docker deployment, and active production cloud hosting, the platform is verified, robust, and production-ready.